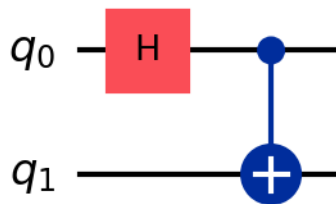


Qiskit v2.x Practice Quiz

Questions

1. What state does the circuit below prepare?



- A. $|00\rangle$
B. $|11\rangle$
C. $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
D. $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
2. Which of these are OpenQASM 3 features? *(Select all that apply)*
- A. while loops
B. angle
C. try/catch
D. barrier
3. What does increasing the number of shots usually improve?
- A. Sampling precision
B. Circuit depth
C. Qubit connectivity
D. Gate fidelity automatically
4. What does the Bloch sphere represent?
- A. Only classical bits
B. Single-qubit pure states
C. Only two-qubit entangled states
D. Classical probability distributions only

5. What does layout refer to in transpilation?
- Mapping virtual qubits to physical qubits
 - Choosing the number of shots
 - Drawing the circuit
 - Choosing classical register names
6. Which code correctly creates a two-qubit quantum circuit?
- QuantumCircuit()
 - QuantumCircuit(2)
 - Circuit(2)
 - QuantumRegister.create(2)
7. Which one of the following images is the output from the code below:

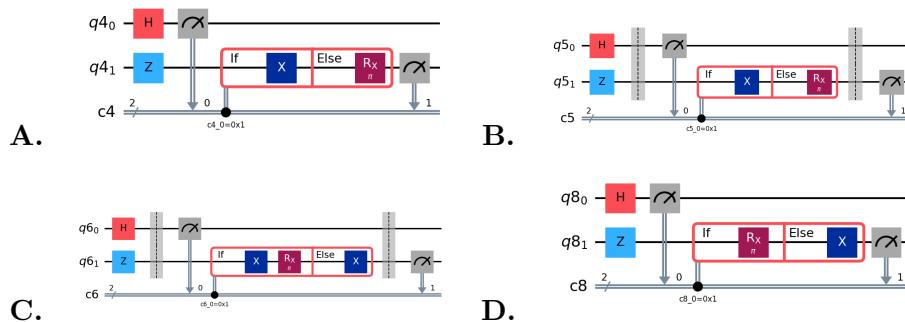
```

from qiskit import *
from math import pi

qubits = QuantumRegister(2)
clbits = ClassicalRegister(2)
qc = QuantumCircuit(qubits, clbits)
(q0, q1) = qubits
(c1, c2) = clbits

qc.h(0)
qc.z(1)
qc.barrier()
qc.measure(q0, c1)
with qc.if_test((c1,1)) as else_:
    qc.x(q1)
with else_:
    qc.rx(pi, q1)
qc.barrier()
qc.measure(q1, c2)
qc.draw(output="mpl", plot_barriers=False)

```



8. What is the purpose of transpilation?
- To convert a circuit into backend-compatible instructions

- B. To measure all qubits automatically
 - C. To create a statevector
 - D. To define an observable
9. Which gate maps $|0\rangle$ to $|+\rangle$?
- A. X
 - B. Z
 - C. H
 - D. CX
10. Which object is commonly used to represent Pauli observables in Qiskit?
- A. QuantumCircuit
 - B. SparsePauliOp
 - C. ClassicalRegister
 - D. PassManager
11. Why may SWAP gates be inserted during transpilation?
- A. To reset qubits
 - B. To satisfy hardware connectivity constraints
 - C. To increase shot count
 - D. To remove all noise
12. What does an X gate do to $|0\rangle$?
- A. Creates $|+\rangle$
 - B. Maps it to $|1\rangle$
 - C. Measures the qubit
 - D. Adds a global phase only
13. What does the Estimator primitive primarily compute?
- A. Bitstring counts
 - B. Expectation values of observables
 - C. Circuit diagrams
 - D. Backend coupling maps
14. Which visualization is commonly used for measurement counts?
- A. Histogram
 - B. Bloch sphere only
 - C. Circuit drawer only
 - D. Density matrix table only
15. What is the result of applying Z to $|1\rangle$?

- A. $|0\rangle$
 - B. $|1\rangle$
 - C. $-|1\rangle$
 - D. $|+\rangle$
16. Which OpenQASM 3 variable declaration corresponds to Qiskit's ClassicalRegister declaration?
- A. bit
 - B. clreq[]
 - C. bit[]
 - D. clbit[]
17. What does the following circuit prepare?
- ```
qc = QuantumCircuit(2)
qc.h(0)
qc.cx(0, 1)
```
- A.  $|00\rangle$
  - B.  $|11\rangle$
  - C.  $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$
  - D.  $\frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$
18. Which state lies on the positive X axis of the Bloch sphere?
- A.  $|0\rangle$
  - B.  $|1\rangle$
  - C.  $|+\rangle$
  - D.  $|-\rangle$
19. When is Sampler more appropriate than Estimator?
- A. When you need bitstring measurement outcomes
  - B. When you need expectation values of observables
  - C. When you need to draw a circuit
  - D. When you need to create a backend
20. What does the Sampler primitive primarily return?
- A. Expectation values
  - B. Measurement samples or bitstring results
  - C. Only statevectors
  - D. Only transpiled circuits

## **Answer Key**

1. S3-004: C
2. S8-005: A, B, D
3. S7-001: A
4. S2-001: B
5. S4-003: A
6. S3-001: B
7. S3-003: A
8. S4-001: A
9. S1-002: C
10. S6-002: B
11. S4-002: B
12. S1-001: B
13. S6-001: B
14. S7-002: A
15. S1-003: C
16. S8-003: C
17. S3-002: C
18. S2-002: C
19. S5-002: A
20. S5-001: B